

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Reviewed and refined **model-generated video annotations** for hand movements and object interactions, validating **action timing boundaries** and correcting label atomicity across 2,000+ video segments using **web-based annotation tools** to ensure **95% accuracy** against project standards.
- Assessed and edited video captions for **single, independent action** descriptions, verifying **left versus right hand identification** in all annotated segments and flagging **low-quality, ambiguous, or incomplete labels** for rework, maintaining **98% labeling consistency** across review pipelines.
- Applied **keen visual discrimination skills** and **high attention to detail** to analyze repetitive video content during extended review sessions, delivering **reliable quality** and meeting **daily throughput targets** of 40+ videos reviewed while adhering to **structured annotation rubrics**.
- Provided **actionable feedback** to enhance annotation models through **clear written and verbal communication**, collaborating with team leads to resolve **edge cases** and document corrective guidance in annotation training materials.
- Operated **time-coded video platforms** and **annotation software** with proficiency, troubleshooted video playback quality issues leveraging **high-speed internet** and optimized computer hardware, and worked productively in autonomous, remote-work environments to maintain **100% on-time delivery** of quality reports.
- Demonstrated **process adherence** and followed **detailed guidelines** across multi-stage annotation review workflows, incorporated prompt feedback from supervisors within **24 hours**, and maintained comprehensive audit trails and quality logs to support continuous model improvement and regulatory traceability.

SKILLS

Video Review & Annotation – model-generated video annotation review, action timing validation, hand movement assessment, caption atomicity verification, left/right hand identification, low-quality label flagging, annotation rubric adherence.

Annotation Tools & Platforms – web-based annotation tools, time-coded video platforms, CVAT, Labelbox, Scale, V7, Encord, video playback optimization.

Quality Assurance & Labeling – visual discrimination, attention to detail, label accuracy verification, annotation consistency, edge case resolution, quality audit trails, feedback incorporation.

Communication & Collaboration – written and verbal communication, actionable feedback provision, team lead collaboration, guideline documentation, training material development.

Domain Expertise – occupational therapy observation principles, kinesiology motion analysis, sports movement tracking, sign language interpretation, human movement recognition.

Technical & Remote Work – high-speed internet optimization, video platform troubleshooting, autonomous task management, daily throughput tracking, process logging.

Data Analysis & Reporting – Power BI, Tableau, data quality metrics, statistical accuracy validation, performance dashboarding, SQL reporting.

EDUCATION AND TRAINING

PROJECTS

Human Motion & Action Annotation Quality Framework (HCL Technologies)

- Designed and implemented a **video annotation quality assurance pipeline** that reviewed **3,000+ hand-movement and object-interaction video clips**, validating **action timing boundaries**, **hand laterality (left/right)** correctness, and **caption atomicity** across multi-stage review workflows, achieving **96% first-pass accuracy** and reducing rework cycles by **35%**.
- Developed **annotation rubric documentation** and **structured review guidelines** covering edge cases, ambiguous label detection, and low-quality segment flagging; trained **5 junior annotators** on quality standards using clear written and verbal feedback protocols, resulting in **consistent 94%+ team-wide accuracy**.
- Built **Power BI performance dashboards** tracking real-time annotation metrics—daily throughput (40+ videos/reviewer), accuracy rates, rework frequency, and process adherence—enabling supervisors to identify bottlenecks and optimize review cycles, reducing turnaround time by **28%** over 6 months.
- Collaborated with annotation model development teams to classify and document **recurring edge cases and annotation failure modes**, providing **actionable feedback** that informed model retraining and improved downstream model-generated caption quality by **18%**.
- Maintained comprehensive audit trails and quality logs across **web-based annotation tools** and **time-coded video platforms**, ensuring regulatory traceability and enabling root-cause analysis of labeling discrepancies; automated compliance reporting using **Python scripts** to generate weekly quality summaries.

Video-Based Occupational Therapy Motion Recognition Study (MSc Data Analytics, NCI)

- Analyzed **400+ annotated video segments** of human hand and object-interaction movements in occupational therapy settings, applying **visual discrimination techniques** and **statistical hypothesis testing** to validate timing annotations and movement classification accuracy using **R / RStudio** and **Python (pandas, NumPy, scikit-learn)**.
- Created **interactive Tableau dashboards** and **Power BI reports** visualizing hand laterality distribution, action timing variance, and label quality metrics across 8 therapists; communicated findings to clinical stakeholders using **clear written and verbal communication** to inform annotation best practices.
- Identified **12 recurring edge cases** in video annotation (ambiguous hand occlusion, rapid action transitions, incomplete segments) and documented structured resolution protocols; authored **detailed annotation guidelines** reducing labeling ambiguity by **22%** in subsequent cohorts.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)